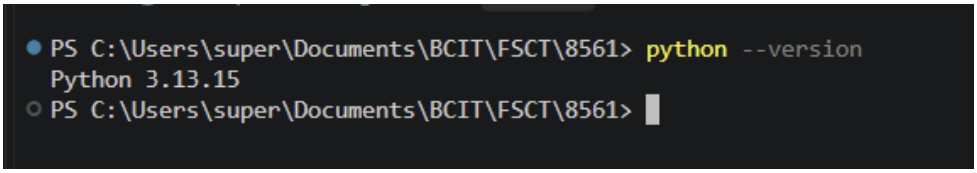
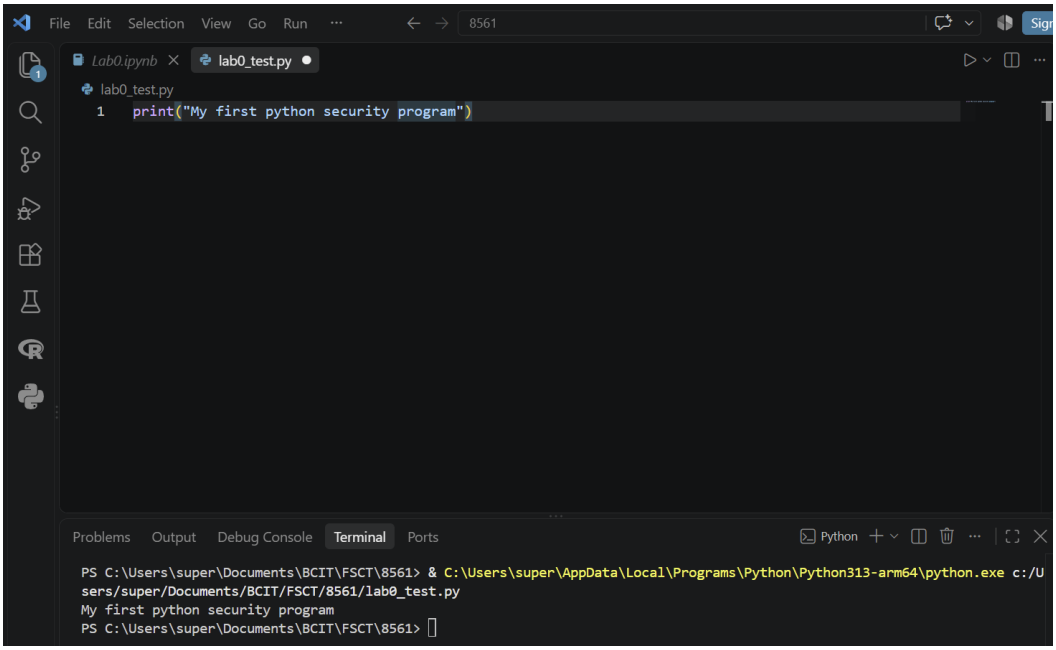
Braeden Venne

Lab 0

Part 0 – Verify Python is Installed



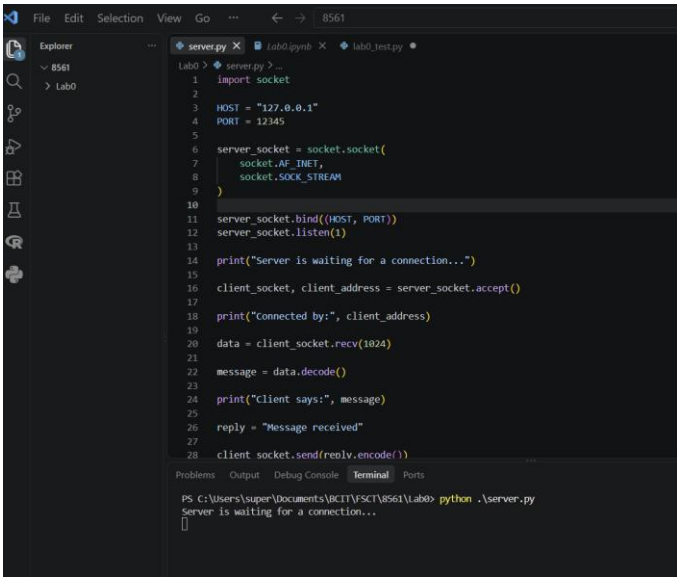
Part 1 – First Python Program



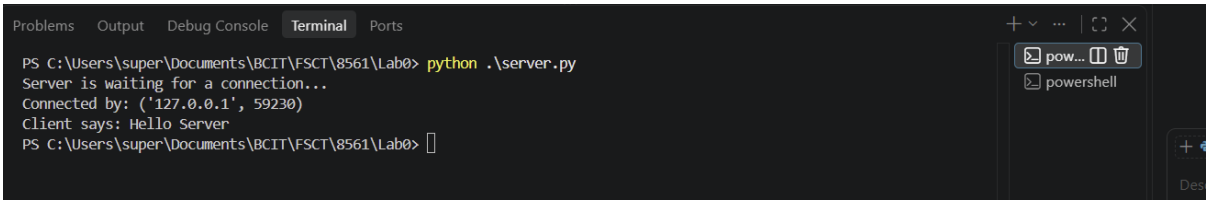
Part 2 – Understanding Sockets

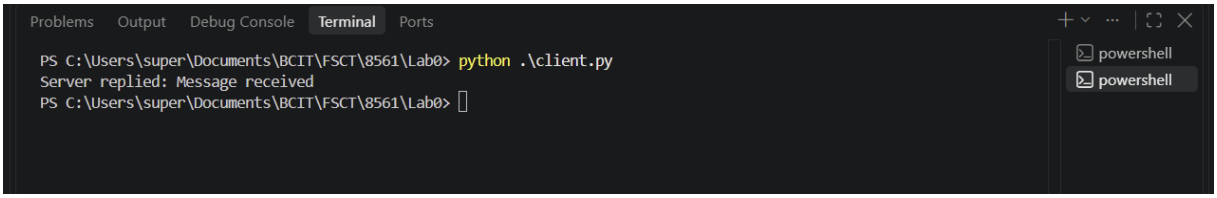
1. What is a socket?
   - A socket is an endpoint that is used by an application to communicate with other programs or computers over a network.
2. What does AF_INET mean?
   - AF_INET refers to the IPv4 address family, it specifies that a socket will utilize IPv4 addresses for communication.
3. What does SOCK_STREAM mean?
   - SOCK_STREAM specifies that the socket will use TCP, which is connection-oriented, bidirectional, and reliable.

Part 4 – Create a Basic TCP Server



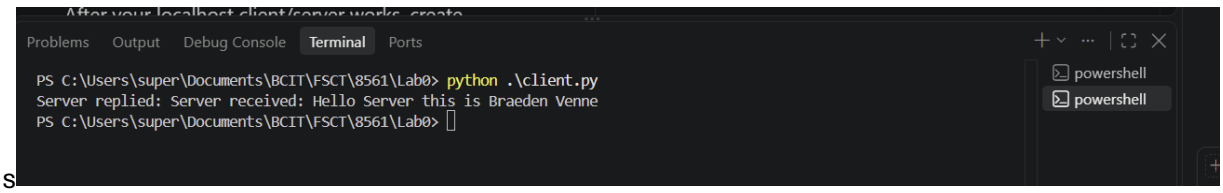
Part 5 – Create a Basic TCP Client





Part 6 – Modify the Client and Server





S

Braeden Venne

Part 7 – Connect to a Public Web Server

raw_socket > http_client.py > ...

```python
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("example.com", 80))

request = (
    "GET / HTTP/1.1\r\n"
    "Host: example.com\r\n"
    "Connection: close\r\n"
    "\r\n"
)

client.send(request.encode())

response = b""
while True:
    data = client.recv(4096)
    if not data:
        break
    response += data

print(response.decode(errors="replace"))
client.close()
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab0> python .\http_client.py
HTTP/1.1 200 OK
Date: Thu, 10 Sep 2026 16:19:24 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: close
Server: cloudflare
Last-Modified: Wed, 02 Sep 2026 22:14:26 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 11221
cf-cache-status: HIT
CF-RAY: a38fc10c2c9fc70b-YVR

22f
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, in
itial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:vi
sited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoi
d use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

0


PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab0>
```

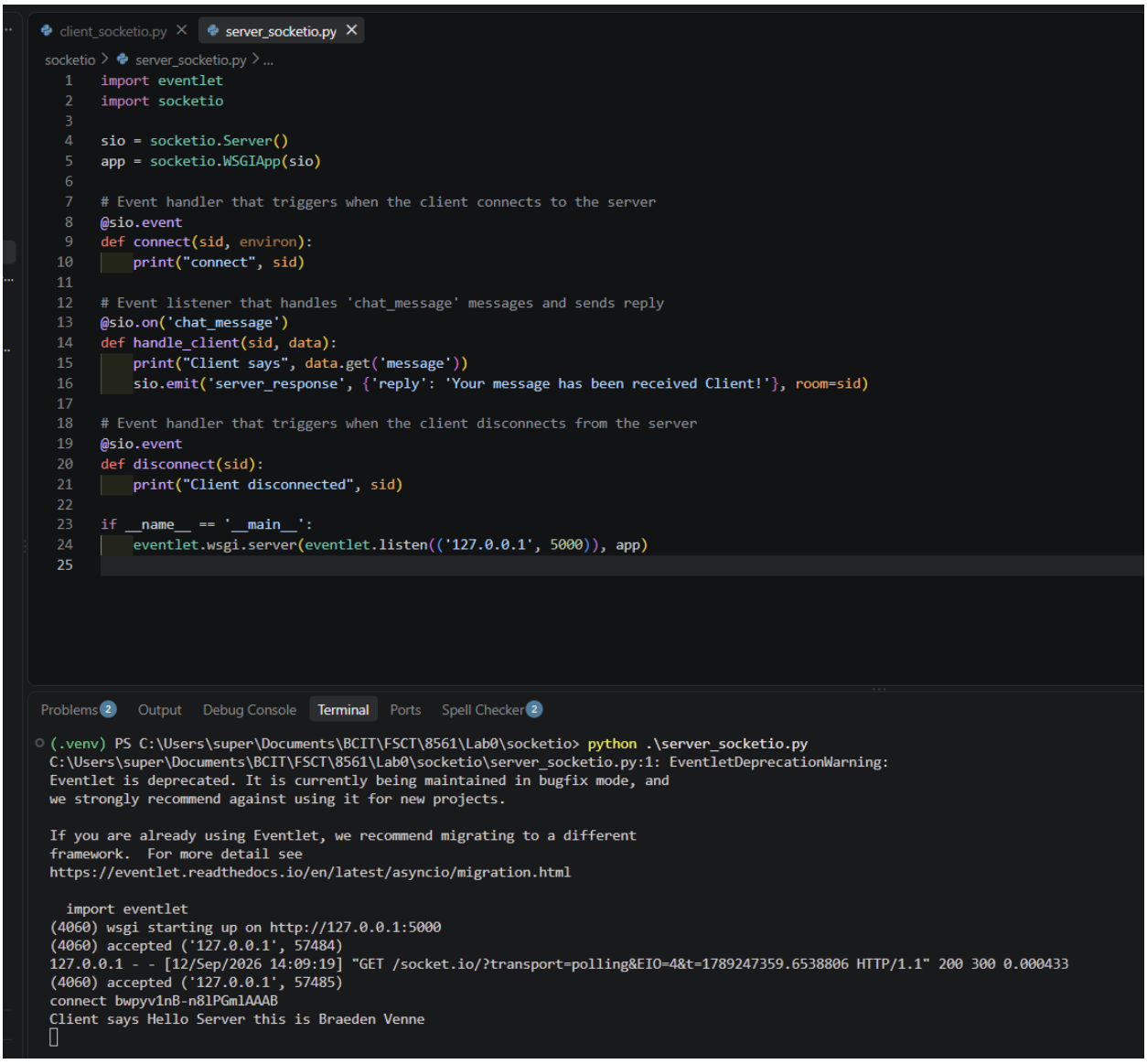
powershell
powershell

1. Which host did the client connect to?
   - The host is example.com
2. Which port was used
   - The port used was port 80
3. Why is port 80 relevant
   - Port 80 is relevant because it is reserved for unencrypted HTTP web traffic.
4. What did the server send back?
   - It sent the response status (HTTP/1.1 200 OK) which indicates the request was successful. It also sent response headers and the raw HTPML content from example.com.

Braeden Venne

Part 8 – Challenge – Rebuild the Client-Server Application Using Socket.IO

All code has been referenced directly from https://python-socketio.readthedocs.io/en/stable/api.html



client_socketio.py | server_socketio.py

socketio > server_socketio.py > ...

```python
import eventlet
import socketio

sio = socketio.Server()
app = socketio.WSGIApp(sio)

# Event handler that triggers when the client connects to the server
@sio.event
def connect(sid, environ):
    print("connect", sid)

# Event listener that handles 'chat_message' messages and sends reply
@sio.on('chat_message')
def handle_client(sid, data):
    print("Client says", data.get('message'))
    sio.emit('server_response', {'reply': 'Your message has been received Client!'}, room=sid)

# Event handler that triggers when the client disconnects from the server
@sio.event
def disconnect(sid):
    print("Client disconnected", sid)

if __name__ == '__main__':
    eventlet.wsgi.server(eventlet.listen(('127.0.0.1', 5000)), app)
```

Problems 2 | Output | Debug Console | Terminal | Ports | Spell Checker 2

```
(.venv) PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab0\socketio> python .\server_socketio.py
C:\Users\super\Documents\BCIT\FSCT\8561\Lab0\socketio\server_socketio.py:1: EventletDeprecationWarning:
Eventlet is deprecated. It is currently being maintained in bugfix mode, and
we strongly recommend against using it for new projects.

If you are already using Eventlet, we recommend migrating to a different
framework.  For more detail see
https://eventlet.readthedocs.io/en/latest/asyncio/migration.html

  import eventlet
(4060) wsgi starting up on http://127.0.0.1:5000
(4060) accepted ('127.0.0.1', 57484)
127.0.0.1 - - [12/Sep/2026 14:09:19] "GET /socket.io/?transport=polling&EIO=4&t=1789247359.6538806 HTTP/1.1" 200 300 0.000433
(4060) accepted ('127.0.0.1', 57485)
connect bwpyv1nB-n8lPGmlAAAB
Client says Hello Server this is Braeden Venne
```

Braeden Venne

```python
import socketio

sio = socketio.Client()

# Event handler that triggers upon the initial connection event
@sio.event
def connect():
    print("Connection established!")
    message = "Hello Server this is Braeden Venne"
    sio.emit('chat_message', {'message': message})

# Event listener for "server_response" events from the server
@sio.on('server_response')
def handle_response(data):
    # Print server reply
    print("Server replied:", data.get('reply'))

# Event handler that triggers when the client disconnects from the server
@sio.event
def disconnect():
    print('disconnected from server')

if __name__ == '__main__':
    sio.connect('http://127.0.0.1:5000')
    sio.wait()
```

```
(.venv) PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab0\socketio> python .\client_socketio.py
Connection established!
Server replied: Your message has been received Client!
```

Disconnect message:

```
(.venv) PS C:\Users\super\Documents\BCIT\FSCT\8561\Lab0\socketio> python .\server_socketio.py
Eventlet is deprecated. It is currently being maintained in bugfix mode, and
we strongly recommend against using it for new projects.

If you are already using Eventlet, we recommend migrating to a different
framework.  For more detail see
https://eventlet.readthedocs.io/en/latest/asyncio/migration.html

  import eventlet
(4060) wsgi starting up on http://127.0.0.1:5000
(4060) accepted ('127.0.0.1', 57484)
127.0.0.1 - - [12/Sep/2026 14:09:19] "GET /socket.io/?transport=polling&EIO=4&t=1789247359.6538806 HTTP/1.1" 200 300 0.000433
(4060) accepted ('127.0.0.1', 57485)
connect bwpyv1nB-n81PGmlAAAB
Venne
Client disconnected bwpyv1nB-n81PGmlAAAB
/socket.io/?transport=websocket&EIO=4&sid=MPI_-1MMYdDEkFTgAAAA&t=1789247359.6588109 HTTP/1.1" 200 0 63.128030
```

Braeden Venne

Part 8 – Security Reflection

1. What could happen if a server accepts connections from anyone?
   - Bad actors would be able to connect and interact with the server without any restrictions. Attackers could conduct a Denial-of-Service attack by flooding the server with fake connection requests.
2. Is the Lab 0 message encrypted?
   - The messages in Lab 0 are transmitted in plain text using standard TCP/HTTP bytes without encryption present.
3. Could network traffic be observed by another system?
   - Since the traffic is unencrypted plain text, it can easily be observed by another system.
4. Why should servers validate input?
   - It can prevent specific attacks/exploits such as SQL Injection and Cross-Site Scripting. Input validation also helps maintain data integrity, specifically by ensuring that only valid and expected data is accepted.
5. Why might authentication be useful?
   - Authentication is useful because it can verify that the client really is who they claim to be, preventing bad actors from stepping in. Additionally, it ensures that the data from the server is only accessible to verified/authorized users.
6. Suggest two ways to make this application more secure.
   - Implementing TLS and adding Authentication tokens.